	PROFICIENCY TESTING SCHEME PLAN	
	Document no: ASMF 7.2.1.3	
Compiled by:	Sifiso Nyambi - Laboratory Manager	
Approved by:	Nonhlanhla Msibi - Operations Manager	

Proficiency Testing (PT) Scheme Plan

Reference: ISO/IEC 17043:2023 Clause 7.2.1.3 | ISO/IEC 17025:2017 Clauses 7.2, 7.6, 7.8 | SANAS R80

Personnel involved

The personnel involved include the Laboratory Analysts, Laboratory Supervisor, Laboratory Manager, and Operations Manager. Each plays a defined role in planning, sample preparation, quality control, and reporting.

External providers

All activities for the production, packaging, verification, and CRM production are conducted internally. No outsourced or subcontracted work is used.

Participation criteria

Participation is open to laboratories that are competent and/or accredited in accordance with ISO/IEC 17025 and apply internationally recognized coal analysis standards as outlined in the PT instruction form.

Number and type of expected participants

Participation is expected from ISO 17025 accredited coal laboratories, typically 10–25 labs per round, including national and regional testing facilities.

Description of activities and results to be reported

Participants will analyse and report results for: ISO Ash (ISO 1171), ISO Volatile Matter (ISO 562), ISO Inherent Moisture (SANS 5925), Quick Ash, Total Sulphur(ASTM D4239), Hydrogen, Total Carbon, Calorific Value(ISO 1928), Ash Fusion Temperature(ISO 540 reducing atmospheres only), Hardgrove Grindability Index(ISO 5074), Abrasive Index(Eskom Method), Total Moisture(ISO 589).

Expected range of values

Ash: 10–40%; Volatile Matter: 10–35%; Inherent Moisture: Non-reproducible (sample dependent); Other analytes within typical South African coal ranges based on CRM characterization data.

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Major sources of error

Sampling: ~85% potential error (controlled by 4-hour drum homogenization). Analysis: ~5% (inter-laboratory variance, instrument calibration). Storage: monitored monthly; stability samples analyzed for any drift occurrence.

Production, Quality Control, Storage and Distribution

Production: PT items produced from homogenised coal sample, blended 4 hours. Homogeneity verified via triplicate analysis. QC: Pre-distribution QC testing with in-house CRMs traceable to SANAS-accredited calibrations.
Storage: airtight containers ≤25°C. Certified reference materials are stored in a temperature-controlled and humidity-monitored storage area.
Distribution: sealed, labeled, tracked shipments.
Traceability: all instruments calibrated to SANAS-accredited standards. Use of internationally recognized calibration standards(Benzoic Acid)

Collusion and falsification prevention

Single Submission: Participants are only given one submission in the submission links provided
Post-closure Analysis: Evaluation of results conducted only after submission deadline;
Simultaneous Release: Reports issued simultaneously to ensure fairness.

Information supplied and time schedule

Participants receive PT instructions after delivery of PT samples. Reporting links follow between 6th and 15th of every participating month, and timelines included in the PT instruction form.

Schedule: Dispatch period no later than 10th of every participating month; Submission period no later than 25th (deadline day 28th of each month); Feedback report to participants no later than the 6th of every participating month.

Frequency for continuous PT schemes

PT releases, deadline days and feedback reports are outlined in the PT instruction form provided to the participating laboratories. PT releases are done on a month to month basis for General Analysis. The lab is making continued efforts to address rigid dates for Semi prep, HGI and AI on an ongoing basis.

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Handling, preparation, and disposal

Requirements for handling, preparation and disposal of PT sample are outlined in the PT instruction form provided. It reads as follows: ***Upon receipt, store samples as follows*:**

- **Temperature:** Room temperature should not be greater than 23 °C.

- **Conditions**: Keep sealed, dry, and away from heat or contamination.

- **Analysis:** Condition the sample for 4 hours prior to analysis.

Homogeneity and stability testing

Homogeneity: 10 random units, duplicate analysis, ANOVA per ISO 13528. Stability: 5 of the retained samples are analyzed monthly, compared to baseline reproducibility limits as per parameter.

Standardised reporting format

A link is sent for each PT participant for the submission of their results each month. It has outlined: Unique Lab ID, Lab Name, asks if participant has read PT instruction prior to submission of results, results for each parameter participated in etc.

Statistical analysis

The lab evaluates the results only after the closure of submission of results. Results are treated whilst on dry basis. Removal of outliers follows, with averages, medians, uncertainties, ranges which all gives reference to standards:

- ISO 33405:2024
- ISO/IEC 17025:2017
- ISO 5725 (Parts 1–6): Accuracy of measurement methods
- ASTM/ISO coal testing methods (e.g., ISO 1171, ISO 562, etc.)

Origin, traceability, and uncertainty

Assignment of values is based on the statistically treated consensus mean values assigned for the PT, which the CRM will be derived from. Assigned values based on in-house labs

traceable to ISO standards. Uncertainty includes uncertainty of characterization, homogeneity, and stability.

Treatment of different methods

Separate data analysis for each method. Combined only where statistical equivalence demonstrated.

Evaluation criteria

The acceptance criteria for results will be based on z-scores wherein the results must lie in the range ±3 in each parameter tested. Monitoring and detection of biases within results is done with the 5 latest z-score points of each parameter.

Data and information returned

Each participant receives homogeneity study, stability study, results submitted by each participant (only unique number appears and not names of laboratories) dry base results table with Z scores, individual z-score graph for each parameter, anonymized(Lab ID only) group report, assigned values, trend data graphs, and conclusions.

Publication and confidentiality

Publication of results is kept confidential with the use of unique Lab IDs only known to participants and the organization. Individual data is disclosed only to individual participants report and only by authorization or regulatory requirement.

Actions for lost, delayed, or damaged items

Replacement issued within 5 days if loss reported within 72h. Late results accepted with proof of dispatch. Non-returns recorded as non-participation. See complaints procedure and material and storage of materials for thorough explanations.